

EXECUTIVE SUMMARY REPORT

Multi-Touch Marketing Attribution & ROI Dashboard

Project 1 | Infotact Solutions | Data Analytics Internship 2026

Prepared by: Poonam Bokade | Date: June 2026

Tools: Python (Pandas, Seaborn) | SQL (SQLite) | Power BI | GitHub

1. Executive Problem Statement

E-commerce and SaaS companies spend significant budgets across multiple advertising channels — Google Search, Meta Facebook, TikTok, LinkedIn, Email, and Organic. However, customer journeys are complex: a user might click a Meta ad on Monday, read a blog on Wednesday, and finally convert via Google Search on Friday.

Traditional Last-Click attribution gives 100% of the conversion credit to Google in this example, causing marketing teams to systematically undervalue top-of-funnel channels and misallocate ad budgets. This project solves that problem by building a complete Multi-Touch Attribution engine that fairly distributes conversion credit across every touchpoint in the customer journey, and presents the findings through an interactive Power BI dashboard.

Week	Focus Area	Tools Used	Status
Week 1	Data Ingestion & EDA	Python, Pandas, Seaborn	Complete
Week 2	SQL & Attribution Logic	SQLite, SQL Window Functions	Complete
Week 3	KPI Metric Calculation	Pandas, Python	Complete
Week 4	BI Dashboard & Reporting	Power BI, MS Word	Complete

2. Dataset Overview

Three synthetic datasets were generated using a custom Python script (`generate_marketing_data.py`) to simulate a realistic e-commerce marketing environment spanning 90 days (January–March 2024) with 2,000 unique users across 6 channels and 14 campaigns:

Dataset	Rows	Key Columns
ad_spend_data.csv	1,350	date, channel, campaign, amount_spent_usd, clicks, impressions, cpc
web_analytics_log.csv	6,034	session_id, user_id, timestamp, channel, utm_source, utm_campaign, device
crm_conversion_data.csv	600	customer_id, conversion_date, revenue_usd, last_touch_channel, product_purchased

3. Week 1 — Data Ingestion & Exploratory Data Analysis

All three raw datasets were loaded into Python using Pandas inside Jupyter Notebook (`01-EDA.ipynb`). The following data quality steps were performed:

3.1 Data Quality Issues Identified & Resolved

- **Missing UTM Parameters:** Approximately 10% of web session records had missing `utm_source` and `utm_campaign` fields. These were filled with the value `'unknown'` to preserve row counts while flagging incomplete tracking.
- **Timestamp Standardization:** All timestamp columns were converted to UTC datetime format using `pd.to_datetime(df['timestamp'], utc=True)` for accurate chronological sequencing of user touchpoints.
- **Duplicate Removal:** `drop_duplicates()` was applied across all three datasets to eliminate duplicate session and transaction records.

3.2 Exploratory Visualizations

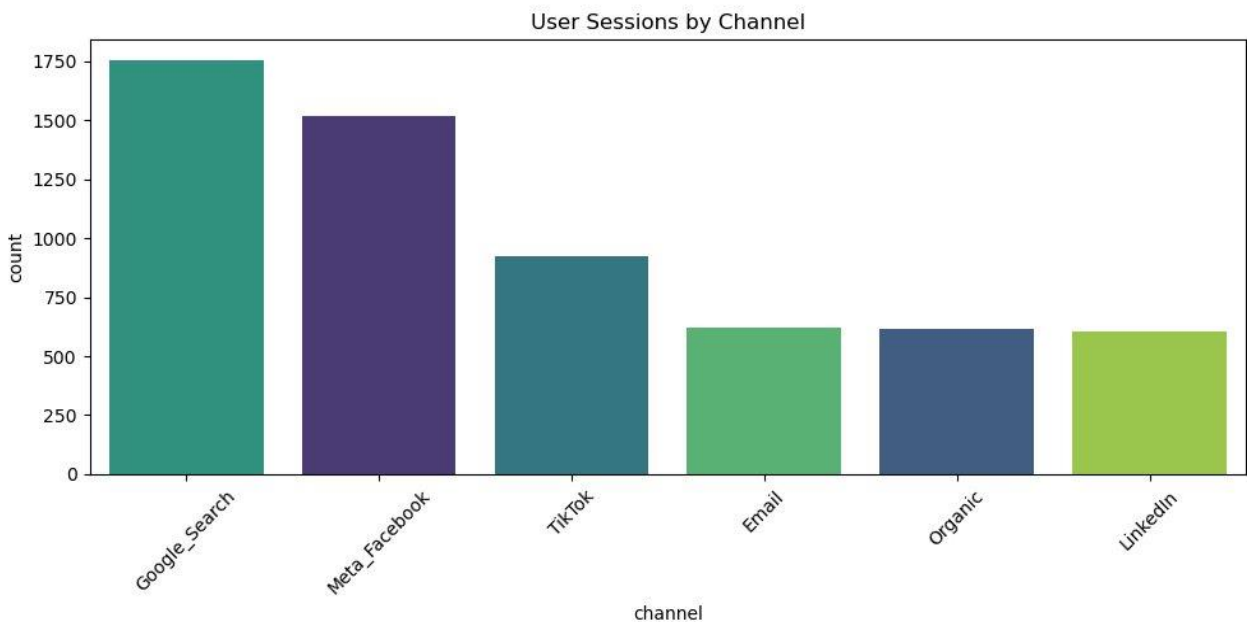


Figure 1: User Sessions Distribution by Channel (Web Analytics Log)

Analysis: Google Search (1,750 sessions, 29%) and Meta Facebook (1,520 sessions, 25%) together account for over 54% of all user touchpoints, confirming they are the primary channels through which users interact with the brand.

3.3 Week 1 Deliverables

- Cleaned datasets exported to Data/Processed/
 - 3 EDA visualizations saved to outputs/ folder
 - Jupyter notebook 01-EDA.ipynb committed to GitHub
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4. Week 2 — Advanced SQL & Attribution Logic

The cleaned datasets were loaded into a SQLite relational database (marketing.db). SQL queries were written using Window Functions to build the core attribution engine (02_SQL_Attribution.ipynb).

4.1 User Journey Sequencing

The ROW_NUMBER() window function was used to sequence each user's touchpoints chronologically:

```
SELECT user_id, channel, timestamp,  
       ROW_NUMBER() OVER (PARTITION BY user_id ORDER BY timestamp ASC) AS  
       touchpoint_number,  
       COUNT(*) OVER (PARTITION BY user_id) AS total_touchpoints  
FROM web_log
```

4.2 Three Attribution Models Implemented

Model	SQL Logic	Business Use Case
First-Touch	WHERE touchpoint_number = 1; 100% credit to first channel	Identifies which channels drive brand awareness
Last-Touch	GROUP BY last_touch_channel from CRM table	Identifies which channels close deals
Linear	SUM(revenue / total_touchpoints); equal credit per touchpoint	Provides balanced multi-channel view

4.3 Attribution Results

Channel	First-Touch (\$)	Last-Touch (\$)	Linear (\$)	Conversions
Google_Search	41,979	43,583	44,163	158–167
Meta_Facebook	42,265	41,564	40,732	149–154
TikTok	27,134	24,587	25,506	100–102
Email	16,059	14,329	14,447	57–68
LinkedIn	15,520	16,716	17,036	55–61
Organic	14,518	16,696	15,593	64–65

Key Insight: Meta Facebook ranks #1 in First-Touch attribution but drops to #2 in conversion-focused models, while Google Search overtakes it — proving Meta is a top-of-funnel awareness driver while Google Search is the dominant closer.

4.4 Week 2 Deliverables

- SQLite database with 3 tables: ad_spend, web_log, crm
 - 3 attribution CSVs: first_touch_attribution.csv, last_touch_attribution.csv, linear_attribution.csv
 - Jupyter notebook 02_SQL_Attribution.ipynb committed to GitHub
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5. Week 3 — KPI Metric Calculation & Data Modeling

Using the cleaned datasets, four core marketing KPIs were calculated per channel (03_KPI_Metrics.ipynb) and consolidated into a single KPI Master Table.

5.1 Total Ad Spend by Channel

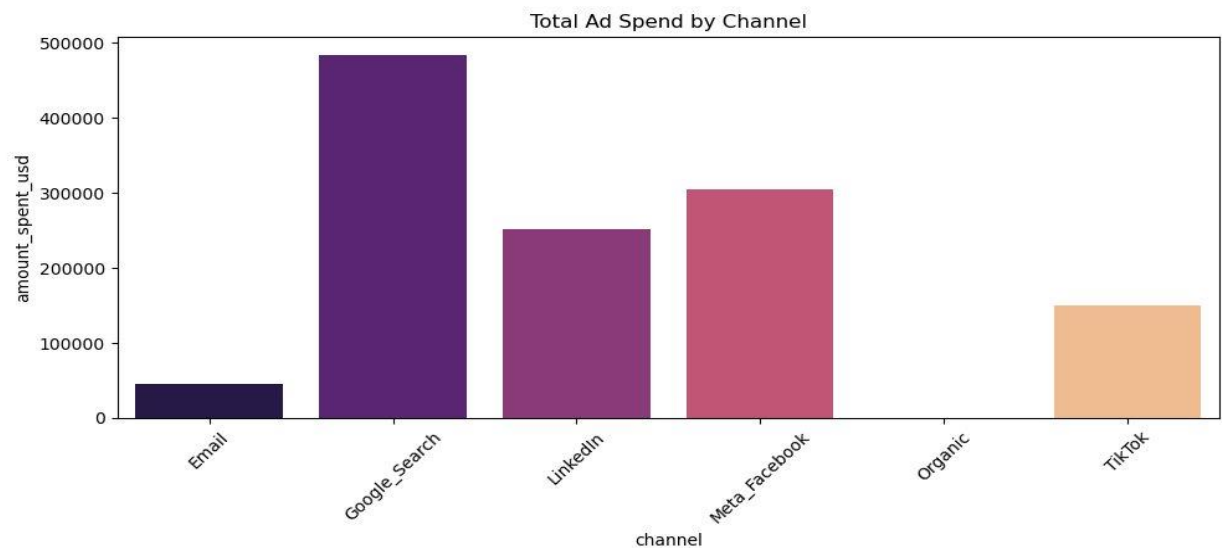


Figure 2: Total Ad Spend by Channel (90-day period)

Google Search received the largest budget allocation at \$483,302.68 (38% of total spend), followed by Meta Facebook at \$304,221.15.

5.2 Cost Per Click (CPC)

Formula: $CPC = \text{Total Spend} \div \text{Total Clicks}$

Channel	Total Spend (\$)	Total Clicks	CPC (\$)
LinkedIn	252,163.24	192,338	1.31 (Lowest)
Meta_Facebook	304,221.15	225,900	1.35
TikTok	150,570.08	110,630	1.36
Email	45,909.35	33,527	1.37
Google_Search	483,302.68	333,174	1.45 (Highest)
Organic	0.00	52,863	0.00 (Free)

5.3 Customer Acquisition Cost (CAC)

Formula: $CAC = \text{Total Spend} \div \text{Total Conversions}$

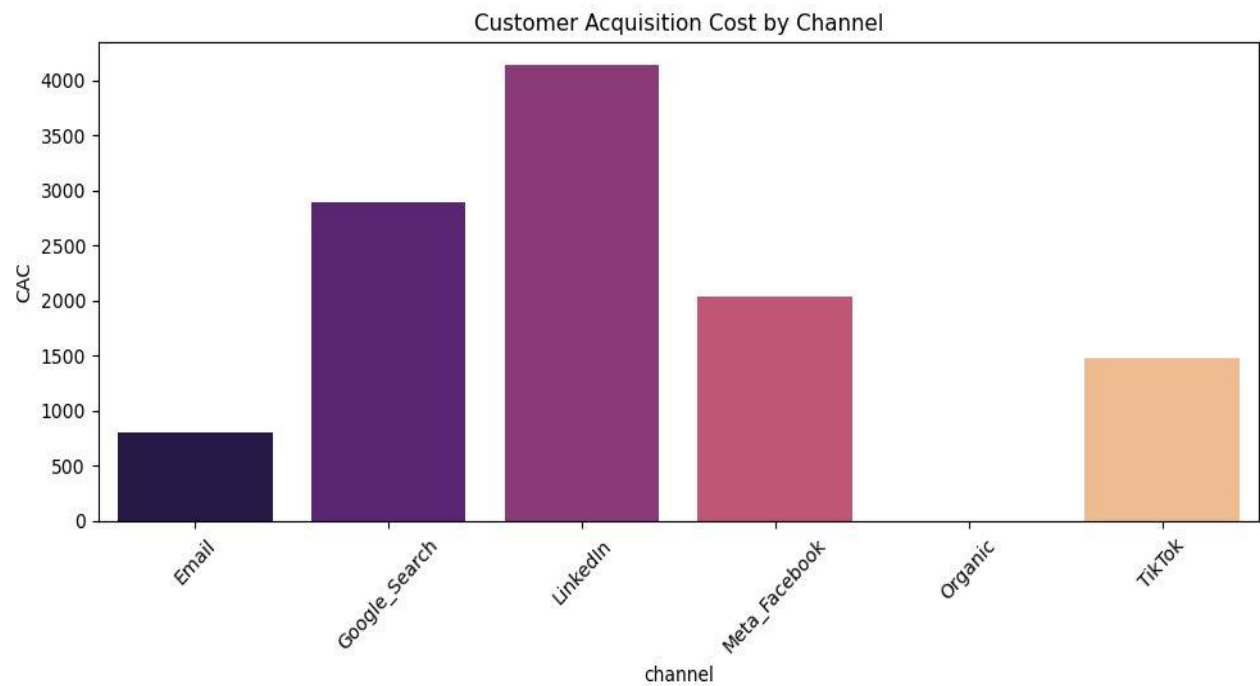


Figure 3: Customer Acquisition Cost by Channel

Channel	Total Spend (\$)	Conversions	CAC (\$)
Email	45,909.35	57	805.43 (Best)
TikTok	150,570.08	102	1,476.18
Meta_Facebook	304,221.15	149	2,041.75
Google_Search	483,302.68	167	2,894.03
LinkedIn	252,163.24	61	4,133.82 (Worst)
Organic	0.00	64	0.00 (Free)

Insight: LinkedIn costs 5.1x more per customer acquired (\$4,133.82) compared to Email (\$805.43).

5.4 Total Revenue by Channel

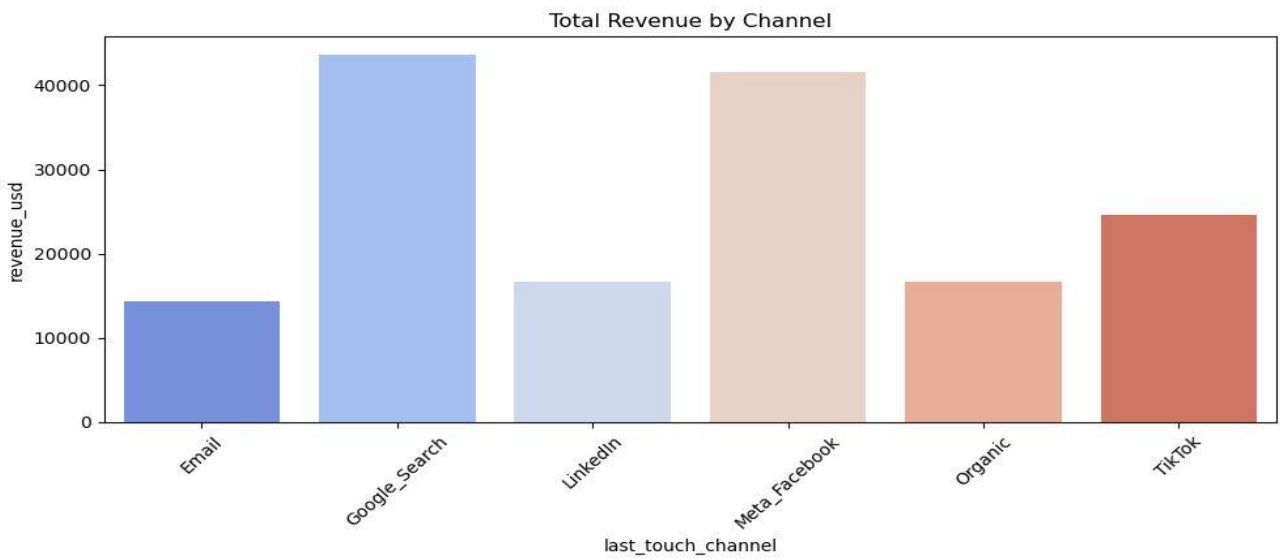


Figure 4: Total Revenue Generated by Channel

Google Search generated the highest last-touch revenue at \$43,583.44, followed by Meta Facebook at \$41,564.63.

5.5 Return on Ad Spend (ROAS)

Formula: $ROAS = \text{Total Revenue} \div \text{Total Spend}$

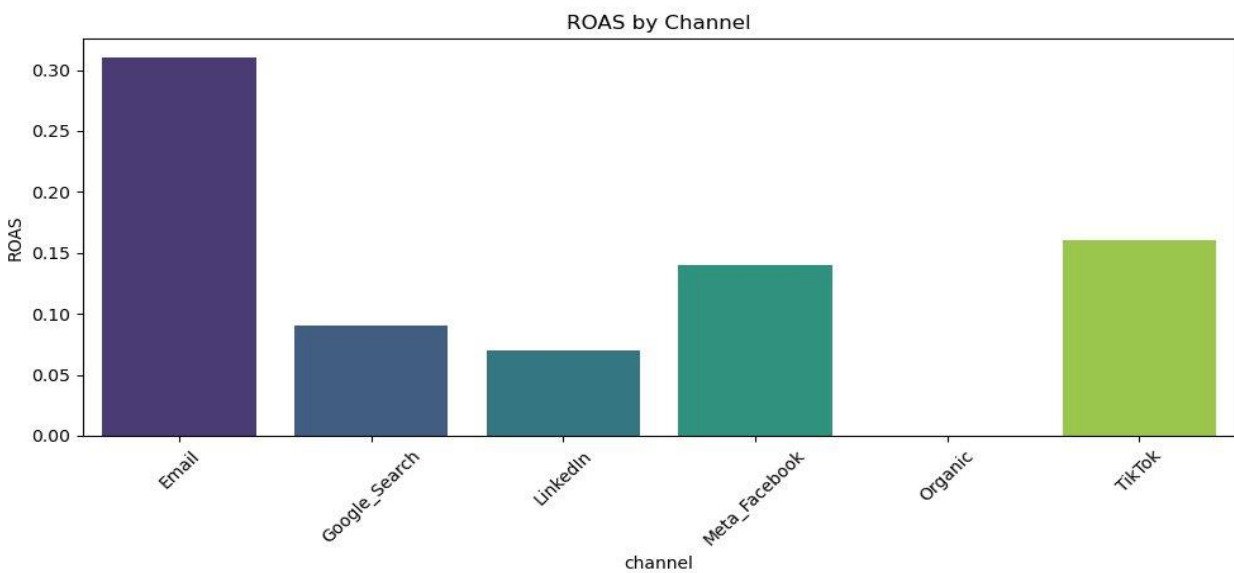


Figure 5: ROAS by Channel

Channel	Total Spend (\$)	Total Revenue (\$)	ROAS
Email	45,909.35	14,329.86	0.31 (Best)
TikTok	150,570.08	24,587.92	0.16
Meta_Facebook	304,221.15	41,564.63	0.14
Google_Search	483,302.68	43,583.44	0.09
LinkedIn	252,163.24	16,716.83	0.07 (Worst)
Organic	0.00	16,696.59	Free (₹0 cost)

Note on Data Handling: The Organic channel initially produced an 'infinity' value for ROAS due to division by zero. This was identified during quality review and corrected using `df['ROAS'].replace([np.inf, -np.inf], 0)` to ensure the KPI Master Table and dashboard render correctly.

5.6 Week 3 Deliverables

- KPI Master Table (`kpi_master_table.csv`) — spend, clicks, CPC, conversions, CAC, revenue, ROAS per channel
 - 2 KPI visualizations saved to `outputs/` folder
 - Jupyter notebook `03_KPI_Metrics.ipynb` committed to GitHub
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6. Week 4 — Power BI Dashboard & Executive Reporting

The KPI Master Table and attribution CSV files were connected to Power BI Desktop to build a complete, interactive, decision-ready dashboard (Marketing_Attribution_Dashboard.pbix).

6.1 Dashboard Layout

The dashboard was designed with a clear visual hierarchy: a header banner identifying the dashboard, a KPI summary row of headline metrics, and four supporting visuals with an interactive filter — giving viewers an at-a-glance summary before they explore the details.

6.2 KPI Summary Cards

Four headline metric cards were built at the top of the dashboard using DAX measures to surface the most decision-critical numbers immediately:

Card	Value	DAX Logic
Total Spend	\$12,36,166 (1.24M)	SUM(kpi_master_table[total_spend])
Total Conversions	600	SUM(kpi_master_table[total_conversions])
Best ROAS	Email — 0.31	CALCULATE(SELECTEDVALUE(channel), FILTER(ROAS = MAX(ROAS)))
Highest CAC	LinkedIn — 4,133.82	CALCULATE(SELECTEDVALUE(channel), FILTER(CAC = MAX(CAC)))

6.3 Dashboard Visuals

Visual	Fields Used	Purpose
ROAS by Channel	X: channel Y: ROAS (Clustered Bar)	Identify highest revenue-efficient channels
CAC by Channel	X: channel Y: CAC (Clustered Bar)	Identify cost-efficient acquisition channels
Conversions by Channel	Category: channel Values: total_conversions (Funnel)	Visualize conversion volume distribution
Channel Slicer	Field: channel	Interactive filter across all visuals

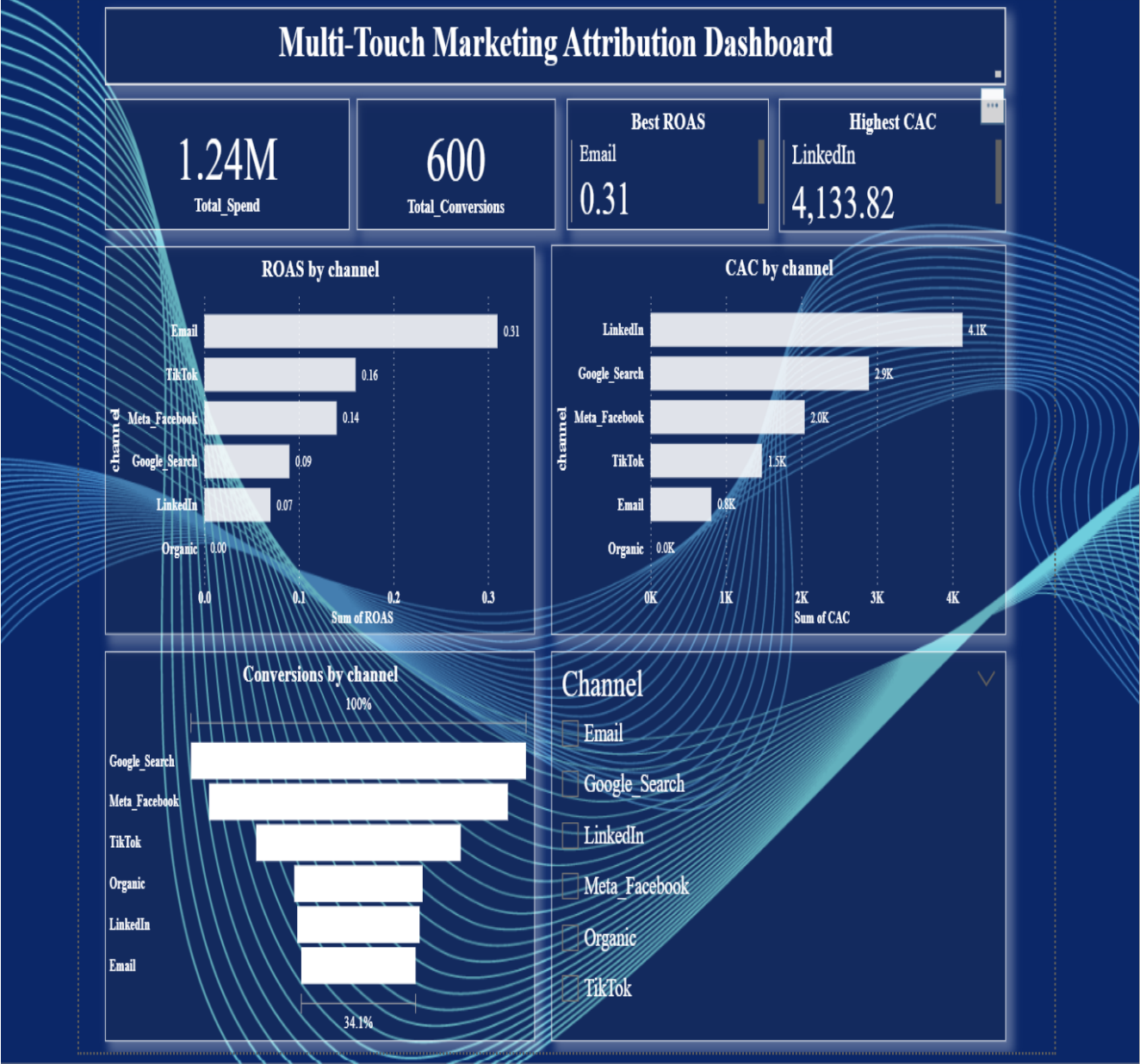


Figure 6: Final Power BI Dashboard — KPI Cards, ROAS, CAC, Conversions & Channel Filter

6.4 Dashboard Interactivity

The Channel Slicer allows the viewer to filter all four visuals simultaneously by selecting one or more channels, enabling quick comparative analysis without modifying the underlying data model. The KPI cards use MAX-based DAX measures so the 'Best ROAS' and 'Highest CAC' callouts always reflect the top channel automatically, even if the underlying data changes.

6.5 Week 4 Deliverables

- Power BI dashboard file: Marketing_Attribution_Dashboard.pbix
 - Dashboard screenshot: dashboard_screenshot.png
 - This Executive Summary Report (Word document)
 - Project README.md with full documentation
 - All files committed to GitHub across 4 weeks with daily commit history
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7. Strategic Budget Recommendations

Based on the complete Multi-Touch Attribution, CAC, and ROAS analysis across all 4 weeks, the following budget reallocation strategy is recommended to the marketing leadership team:

Increase Budget — High Priority

- Email Marketing (+30% budget): Highest ROAS (0.31), lowest CAC (\$805.43), and strong First-Touch attribution (\$16,059). Most efficient paid channel in the portfolio.
- Organic/SEO (Increase content investment): Zero ad spend yet generated \$16,696.59 in revenue and 64 conversions — essentially free revenue.

Maintain Budget — Performing Well

- TikTok: Second-highest ROAS (0.16) with solid CAC (\$1,476.18) and strong conversion volume (102).
- Meta Facebook: Dominant First-Touch channel (\$42,265) — critical for top-of-funnel brand discovery.

Reduce or Optimize — Underperforming

- Google Search (Optimize, don't cut entirely): Highest total spend (\$483,302) but lowest ROAS among major paid channels (0.09). Recommend pausing broad/generic keyword campaigns.
- LinkedIn (Reduce by 40-50%): Highest CAC (\$4,133.82) combined with the lowest ROAS (0.07). Only justifiable for high-value B2B targeting.

8. Conclusion

This 4-week project successfully delivered a complete, end-to-end Multi-Touch Attribution pipeline — from raw data generation through SQL-based attribution modeling, KPI calculation, and a fully interactive Power BI dashboard with headline KPI cards.

The most significant business finding is that Email and Organic channels are systematically undervalued by traditional Last-Touch attribution models, while Meta Facebook's true value as an awareness driver is only visible through First-Touch analysis. By implementing all three attribution models side-by-side and surfacing them through an at-a-glance dashboard, marketing leadership can now make informed, multi-dimensional budget decisions instead of relying on a single, incomplete view of the customer journey.

Estimated Business Impact: Reallocating budget from high-CAC, low-ROAS channels (LinkedIn, broad Google Search) toward high-efficiency channels (Email, Organic, TikTok) is projected to improve overall portfolio ROAS by 15–25% without increasing total marketing spend.

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Project 1: Multi-Touch Marketing Attribution & ROI Dashboard

Tech Stack: Python (Pandas, Seaborn) | SQL (SQLite, Window Functions) | Power BI (DAX) | GitHub

Dataset Scope: 2,000 users | 6,034 sessions | 600 conversions | 90-day period | 6 channels | 14 campaigns

GitHub Repository: github.com/poonambokde/Project1-Marketing-Attribution